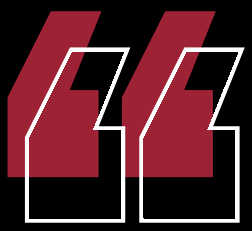

PROFESSIONAL
EDUCATION
Digital Plus+
Programs
Professional Certificate Program in
Digital Transformation



There is not one technology that dominates the current landscape, all of them complement each other to provide data and lead change in companies

Prof. John R. Williams

Index

03 In the Midst of the Digital Transformation

04 About the Certificate

05 Certificate Structure

09 The Learning Experience

10 Key Takeaways

11 Participant Profile

12 Instructors

16 Certificate Outline

17 Methodology and Assessment

18 Impact Project

18 Learning Facilitators

19 Professional Development Webinars

20 Get Your Certification

22 About MIT Professional Education

23 Corporate Courses

24 The Beyond Online Methodology

25 Benefits of Joining the Alumni Community

In the Midst of the Digital Transformation



Our world today is not simply changing. It is in a process of constant evolution, which of course has tremendous effects on business.

Now, cultural diversity abounds, and the key to making an impact depends on having a strong understanding of the latest disruptive technology.

Business leaders must possess a global vision and be able to inspire those around them, a critical skill for the successful implementation of a diverse range of projects.

This transformation, born from technological change, has impacted organizations both on a procedural and cultural level.

About this Certificate



Managers need to adopt an innovative mindset and improve their leadership skills,both for developing and retaining talent as well as for creating high-performance teams that can adapt to any situation.

The Professional Certificate Program in Digital Transformation is MIT Professional Education's response to this need.

This certificate prepares you to **understand the new digital paradigm of today's technological transformation and evolution,** and it also provides an overview of the key technologies driving this change. This certificate takes a practical approach intended to help you become an active part of the digital transformation.

9 months | 8-10 hours a week | Online

Certificate Structure

Core Courses

The Professional Certificate in Digital Transformation offers a holistic vision of many possible **applications in different industries, sectors, and types of organizations**. The following diagram summarizes the formative courses.

In this 9-month online program, participants will fully explore these three courses, plus two electives, to better understand and champion digital transformation.



We live in a highly globalized world. It's important for professionals to adapt to this new culture of continuous innovation to achieve personal and professional success.

Organizations and projects increasingly need the combination of such technical excellence with strong interpersonal and leadership skills. 'Soft skills' are needed in order to emerge strengthened from this environment of constant innovation and:

- Create a culture of driving change.
- Develop a global vision of the projects.
- Train the ability to inspire and motivate people.





Given that our economy is currently undergoing a rapid evolution, professionals need to immerse themselves in technological environments in order to understand the technologies that are driving this change.

Elective Courses

Blockchain connects decentralized systems in a format that is open source, versatile, and unalterably secure without depending on third parties so that everyone can participate directly in the exponential change that is taking g place





Organizations are increasingly more dependent on large information flows, so the demand for safer and faster environments is growing rapidly and exponentially. When the infrastructure that supports technology changes structurally, it is essential to understand this change in order to make strategic decisions about the future of a business or project.

With a focus on product design, manufacturing, and economies of scale in product development, this online program explores how product architecture, platforms, and commonality can help an organization deploy and manage a family of products in an increasingly competitive environment.





Digital platforms and two-sided markets have quickly established themselves as a staple of modern business, presenting new opportunities and challenges alike for engineers, product managers, regulators, and entrepreneurs. However, there continues to be little discussion surrounding why some digital platforms help companies flourish where others have fallen short.

With a focus on product design, manufacturing, and economies of scale in product development, this online program explores how product architecture, platforms, and commonality can help an organization deploy and manage a family of products in an increasingly competitive environment.

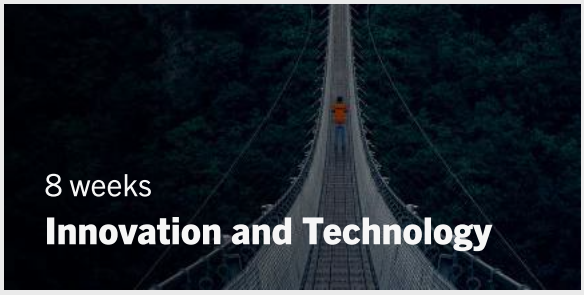




Our daily lives are filled with a barrage of information. Accordingly, the demand for professionals to communicate clearly and effectively is more urgent than ever. More than simply transmitting their ideas, professionals and innovators must now learn to capture and maintain their audiences' attention.

Machine Learning, a branch of Artificial Intelligence, is becoming a fundamental tool for making sound decisions by analyzing large quantities of data and events. Its objective: reduce spaces of uncertainty and arbitrariness through automatic learning and provide organizations and professionals the security needed to make impactful decisions.





8 weeks

Innovation and Technology

In this world marked by tight competition, organizations are required to innovate and develop at the speed of light to remain ahead. This exhilarating environment requires leaders who adapt, continuously learn, and understand that technology is key to take their organizations to a more beneficial, optimized, and connected reality.



10 weeks

Women in Leadership:
Becoming an Agent of Change

Smart manufacturing is the convergence of techniques used in modern data science and artificial intelligence to create processes that can be used in the factory of the future. But why is it necessary today? This technology increases efficiency and eliminates system weaknesses, connecting all organizations and operating systems and ensuring greater productivity, sustainability, and economic growth.

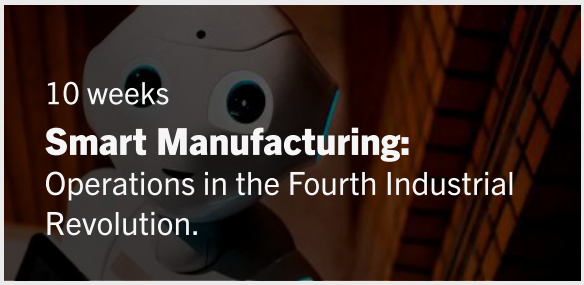
You will explore the systematic process of innovating. In doing so you will examine the role of critical thinking in innovating, design your own toolkit to solve real-world problems, and learn more about the benefits this framework can offer you and your organization.



9 weeks

Management of Technology:
Roadmapping & Development

You will explicitly and honestly learn the challenges female-identifying individuals face in organizations and businesses. This course is primarily forwomen wanting to fully comprehend obstacles that they may face and how to successfully overcome them.



10 weeks

Smart Manufacturing:
Operations in the Fourth Industrial Revolution.

REGISTER

The Learning Experience

At MIT Professional Education we believe in practical applications and encourage participants to implement what they have learned. Through our programs, participants are given the tools to make their learning experience as fulfilling as possible. Participants will have the chance to participate in an engaging and collaborative classroom environment with discussion forums and group activities.

The program is 100% online, and the virtual campus is available anytime, anywhere. It is a space for continuous learning, taking you beyond the limitations of a traditional classroom.

There are no prerequisites for this certificate. The certificate will guide and prepare you to understand all the content and complete the required activities.



Key Takeaways

01 Learn how to combine excellence in the implementation of **technical expertise** with **interpersonal and leadership skills.**

02 Deepen in the **context of globalization** to which we are exposed to be able to operate in this new culture of **continuous innovation.**

03 Discover the five technologies that are transforming our societies and how they are doing it: **Blockchain, Cloud, Artificial Intelligence (AI), Internet of Things (IoT) and Cybersecurity.**

04 Become familiar with **Cloud Computing and DevOps** to analyze the best approach to leveraging their potential for **secure software design and efficient technology operations.**

05 Cultivate a deeper understanding of the **fundamentals of blockchain** and its **practical applications.**

06 Understand the **platform and product family'sconcepts** and be able to **prioritize drivers of modularity** and product platform design to identify new market and product opportunities and **generate revenue growth.**

07 Help others hone their **interpersonal skills in vital areas** such as public speaking, critical thinking, digital communication.



Participant Profile

Professional Certificate Program in Digital Transformation is aimed at various professional profiles. This includes:

Professionals with strong leadership skills who want to have a powerful impact on the success of their organization's innovation processes.

CEOs, general directors, and other functional managers in technical areas whose companies are going through digital transformations and who want to create new business models and optimize their current operations.

CTOs, CInOs, and technical professionals who want to obtain a broader strategic vision about how these key technologies can transform their business operations.

Entrepreneurs of innovative technological startups.

Any professional who wants to understand the benefits and opportunities that cultural diversity, leadership in innovation, and the latest technologies can offer for digital transformation in their organization.

REGISTER

Instructors



Bhaskar Pant

Executive Director of MIT Professional Education

Bhaskar Pant is the executive director of MIT Professional education, the branch of MIT that provides technology professionals access to MIT knowledge. Under his direction, the MIT Professional Education offering has grown considerably,as have corporate relationships and its globalreach, as professionals from more than 120 different countries participate in the classroom, internationals and, recently, also in those launched in online and mixed formats. The courses, which cover such cutting-edge areas as big data, artificial intelligence, and the Internet of Things (IoT), combined with other professional development courses in the fields of innovation and entrepreneurship, have benefited thousands of professionals worldwide.

"Much of innovation and leadership is cultural awareness."

Mr. Pant was part of the "The Future of Education at MIT" team, created by the MIT President, and contributed specifically to the work of the subcommittee that was responsible for "opening the doors of MIT to the world".

Building on Mr. Pant's initiative, whose objective was to expand globally, MIT Professional Education offers short programs in an increasing number of regions in Asia, Latin America, the Middle East, and Europe

Prior to joining MIT, Mr. Pant held important leadership positions in education and corporate companies around the world.



Dr. David Niño

Senior Lecturer in the Daniel J. Riccio Graduate Engineering Leadership Program at MIT

Dr. David Niño teaches in MIT's Daniel J. Riccio Graduate Engineering Leadership Program. Previously, he was a Professor of Engineering Leadership Practice and Management at Rice University. During this tenure, he led the creation of the first four-year certificate in engineering leadership and was director of Rice Leadership, the university's leadership development program.

A former professor of management, Dr. Niño has published on leadership, ethics, corporate culture, and the development of professional management skills. He has conducted research in cutting-edge technology environments and is currently interested in researching how leadership develops among engineers and in engineering organizations.

"Professionals who want to come to MIT and want to learn how to be a more effective and more impactful professional will benefit from this course."

Dr. Niño has taught leadership since 1998 at the undergraduate, master's, doctoral, and executive/professional levels.

He is currently a founding officer of the Leadership Development Division of the American Society for Engineering Education and advises professionals and executives on leadership and development issues.

He holds a Ph.D. in Management from the University of Texas, Austin, where he also studied for his Bachelor of Arts, Bachelor of Business Administration, and Master of Arts.



Dr. Abel Sanchez

Executive Director of MIT's Geospatial Data Center

Dr. Sanchez is the architect of the global network "The Internet of Things" and data analysis platforms for SAP, Ford, Johnson & Johnson, Accenture, Shell, Exxon Mobil, and Altria. In cyber security, he has developed cyber-attack impact analyses for the U.S. Department of Defense and a password firewall for the IARPA.

Dr. Sanchez holds a Ph.D. from the Massachusetts Institute of Technology (MIT) and teaches MIT courses in cybersecurity, engineering, blockchain, and data science. He has been involved in developing educational software for Microsoft and establishing the Accenture Technology Academy. He has produced over 150 educational videos, has 10 years of experience with learning management systems, and has made deployments in the Americas, Asia, and Europe.

"We cannot bet on the artificial intelligence, or the human being separate. We must get the fusion of both."

Involved in the development of educational software for Microsoft and in the establishment of the Accenture Technology academy.

Produced more than 150 educational videos and has ten years of experience with management systems learning.

Made deployments in the Americas, Asia and Europe.



Prof. John R. Williams

Professor of Information Engineering, Civil and Environmental Engineering and Director of MIT Geospatial Data Center

Professor John R. Williams' research focuses on the development and application of computing algorithms in distributed cyberphysical systems. He was director of the AutoID Laboratory, where the Internet of Things was invented.

He is considered, along with Bill Gates and Larry Ellison, one of the 50 most powerful people in "Computer Networks". He is the author and co-author of more than 250 articles in journals and conferences. Professor Williams teaches courses on the basics of programming, modern software development, and the architecture of web, cloud, and blockchain systems.

"Once you climb that first mountain of knowledge, you will be able to teach yourself many things."

Author and co-author of more than 250 articles in magazines and conferences.

Earned a bachelor's degree in Physics from the University of Oxford and a Masters in Physics from UCLA.

arned his doctorate from Swansea University, in the United Kingdom.

Certificate Outline

Course 1
Cultural Awareness for Global Business

- Introduction to Cultural Awareness
- Intercultural Communication: The role of verbal and non-verbal communication

Course 2
Leadership and Innovation

- Why leadership matters in theinnovation process?
- My leadership skills
- Creating and communicating a shared vision
- Analysis of a real project

Course 3
Digital Transformation: From AI and IoT to Cloud, Blockchain, and Cybersecurity

- Blockchain: Introduction and Applications
- Key Concepts of Blockchain
- Cloud
- Artificial Intelligence
- The Internet of Things
- Cybersecurity

Courses 4 and 5 Electives*

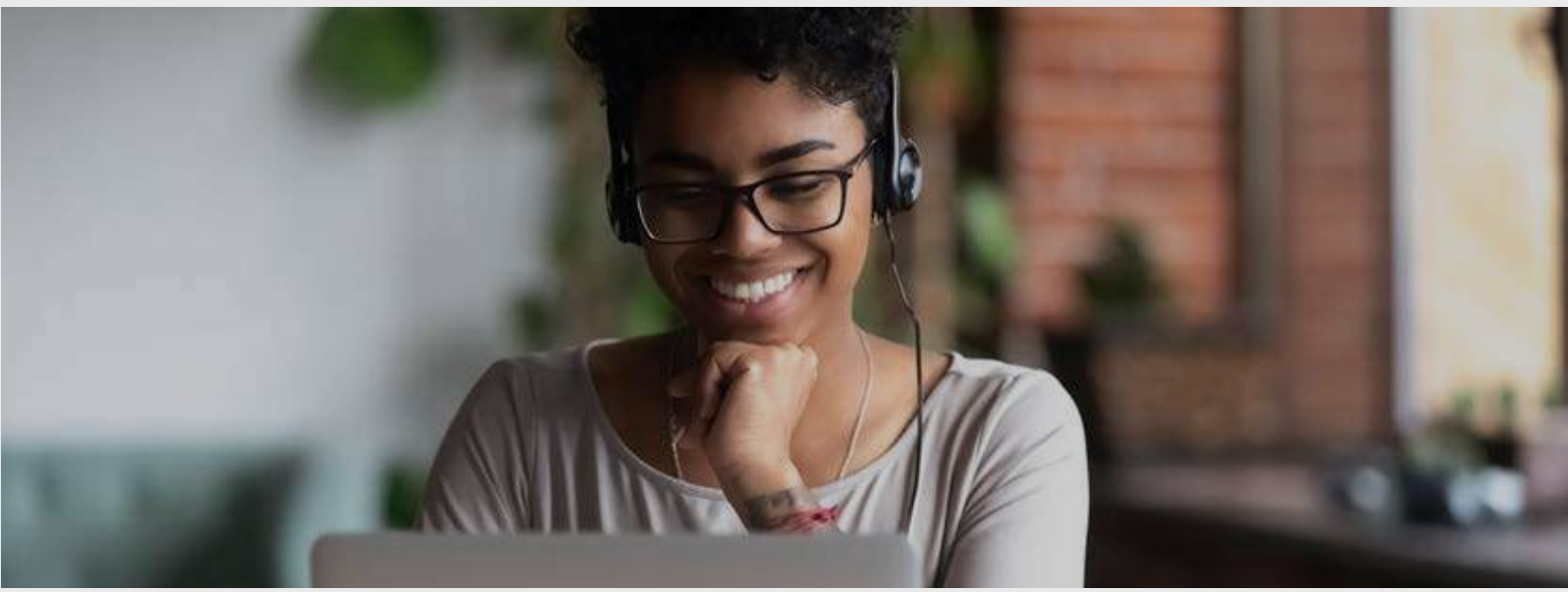
- Blockchain: Disruptive Technology (8 weeks)
- Cloud & DevOps: Continuous Transformation (8 weeks)
- Designing High-Impact Solutions with MITdesignX (8 weeks)
- Designing Product Families: From Strategy to Implementation (8 weeks)
- Digital Platforms: Designing Two-Sided Markets from APIs to Feature Roadmaps (8 weeks)
- Persuasive Communication: Critical Thinking to Enhance your Message (9 weeks)
- Machine Learning: From Data to Decisions (8 weeks)
- Management of Technology: Roadmapping & Development (9 weeks)
- Innovation and Technology (8 weeks)

REGISTER

Methodology and Assessment

All the courses included in the Professional Certificate in Digital Transformation take place in highly collaborative environments, fostered by discussion forums and group activities. In this practical, results-driven program you will get a hands-on learning experience based on the case method. Plus, you will acquire skills and tools that you can immediately use to benefit your organization and your career. Participants will be guided throughout all courses by expert facilitators who will be on hand to answer any questions and respond to comments.

- Practical activities
- Self-assessment
- Review activities
- Live online sessions: By professors and industry experts who will teach real-life applications and respond to participants' questions
- Additional resources: Videos and recommended readings
- Guided group forums
- Case applications



Impact Project

The Impact Project is a cross-cutting project that runs throughout the Professional Certificate. It is group work, which serves as a culminating academic experience.

This project is designed for the student to reflect on the topics covered throughout the certificate, make a practical application of such knowledge, integrate, and connect the knowledge acquired in each of the courses and perform, together with their working group, a presentation, and defense of the proposed solution to the business challenge posed

In addition to the above, the project has several additional objectives such as enhancing teamwork, verbal communication skills, research skills, planning and the ability to solve complex problems.



Learning Facilitators

The program will be supported by Learning Facilitators. These senior consultants are subject matter experts, with vast industry experience. They will be available to answer questions, to probe to generate deeper discussion, and to facilitate a collaborative, positive learning experience via discussion forums and live video sessions.

Professional Development Webinars

Throughout the professional certificate, participants will be able to participate in professional development webinars with subject matter specialists. Each of these sessions, which will last 1h, will take place in the fourth week of each module, distributed as follows *:

- Fourth week of the program in **Leadership and Innovation.**
- Fourth week of the program in **Digital Transformation: From AI and IoT to Cloud, Blockchain, and Cybersecurity.**
- Fourth week of the program in **Elective Course 1.**
- Fourth week of the program in **Elective Course 2.**

The topics to be covered in these sessions are as follows:

Session One	Personal Branding
Session Two	Communication Skills
Session Three	Negotiation skills
Session Four	Networking and the headhunter ecosystem

* Sessions may change depending on the availability of the speaker and MIT Professional Education's academic portfolio

Certificate of Completion

All participants who successfully complete the **Professional Certificate Program in Digital Transformation** will receive an MIT Professional Education Certificate of Completion. In addition, they will also earn 31.6 Continuing Education Units (MIT CEUs*).

In order to obtain MIT CEUs, participants must complete a required MIT CEU confirmation form. MIT CEUs are calculated based on the number of learning hours in each course.



The Continuing Education Unit (CEU) is defined as 10 contact hours of ongoing learning to indicate the amount of time they have devoted to a non-credit/non-degree professional development program. To understand whether or not these CEUs may be applied toward professional certification, licensing requirements, or other required training or continuing education hours, please consult your training department or licensing authority directly.



Welcome to the gateway

to MIT knowledge and expertise for professionals around the world.

155

Participants from 155 countries

+1600

More than 1,600 online activities

92%

92% of Professional Alumni rated their experience as outstanding

About MIT Professional Education

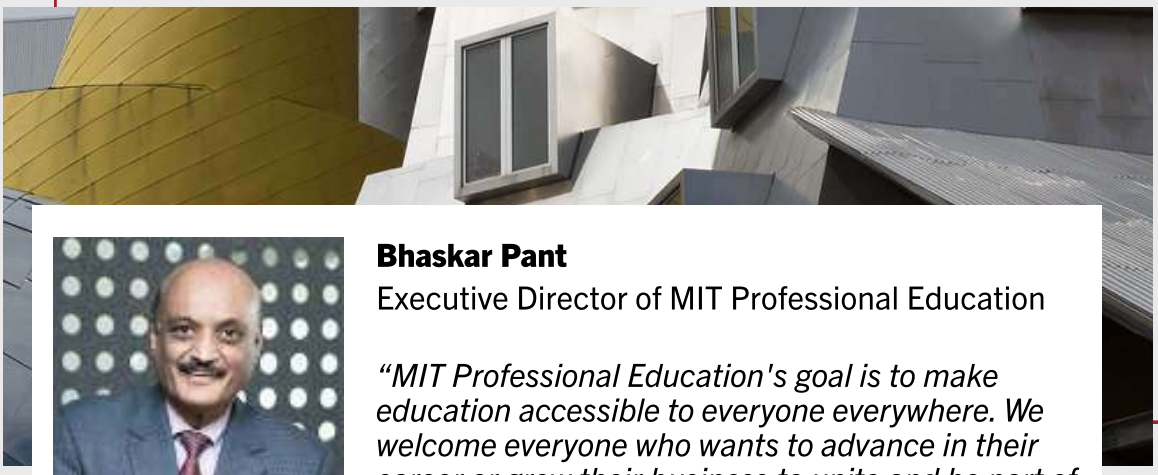
MIT Professional Education fosters the development of innovative leaders equipped to address complex problems globally

MIT Professional Education provides continuing education courses and lifelong learning opportunities for science, engineering, and technology professionals at all levels, from around the world. Worthy of note is that **MIT professors and lecturers lead and teach all MIT Professional Education offerings**. Some of our programs have a long history. Others are relatively new.

Our Digital Plus Programs go beyond online, blending innovative content delivered using the best of online technology and traditional classroom instruction in a flexible, collaborative learning environment.

MIT Professional Education is central to MIT's vision. It fulfills the mandate to connect practitioner-oriented education with industry and incorporate industry feedback and knowledge into MIT education and research.

The Institute is committed to generating, disseminating, and preserving knowledge while working with others to apply this **knowledge for the benefit of humankind.**





Bhaskar Pant
Executive Director of MIT Professional Education

"MIT Professional Education's goal is to make education accessible to everyone everywhere. We welcome everyone who wants to advance in their career or grow their business to unite and be part of the MIT Professional Education all-inclusive community."

Corporate Courses

At MIT Professional Education, we are committed to accompanying organizations as they aspire to provide their professionals with the ongoing education critical for **success in today's competitive environment**. That's why we offer plans for companies who wish to train their professionals with us.

Which program best suits your organization's needs?

1

Standard Program:
Teams or groups of at least 10 people will obtain special conditions in their registration for any of our programs.

2

Custom Programs:
Custom-made to meet the specific needs of each organization and its professionals.

3

Private Cohorts:
Available in Open Enrollment courses with a minimum of participants.

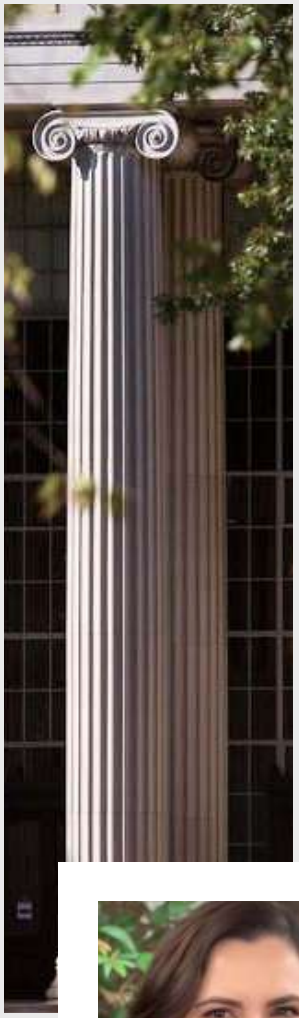


For more information on our Programs for Organizations, please contact us at: **professionalprograms@mit.edu**

The Beyond Online Methodology

MIT Professional Education is **revolutionizing the digital learning experience**. In fact, we are no longer talking about online-only, but rather an **interactive and collaborative learning experience** that's digitally powered: a **Beyond Online experience.**

You will have access to one of the most innovative e-learning platforms that utilizes the latest technology.You will also have an opportunity to meet and work with some of the world's leading subject matter experts in their respective fields.



With MIT Professional Education, you'll have an opportunity to:

- Attend select live webinars with MIT Instructors.
- Collaborate and engage with world-renowned MIT Instructors.
- Learn from expert learning facilitators, guest speakers, and subject matter experts from industries.
- Weekly live webinars from learning facilitators.
- Fulfill your reskilling and upskilling needs in this fast-changing environment.
- Master the technical and human skills necessary for effective leadership.
- Meet and network with other program participants from all around the world.



Clara Piloto
Director of Global Programs at MIT Professional Education

"MIT Professional Education's programs tackle all the technologies that are at the heart of the digital transformation, enabling dedicated professionals to meet new challenges in the Fourth Industrial Revolution and lead change within theirorganizations."

Benefits of Joining the Alumni Community



MIT Professional Education offers **a number of benefits for participants** who successfully complete our courses:

1. **Certificate Award Ceremony at MIT's Campus in Cambridge**, Massachusetts, where participants will receive a physical diploma
2. **Two years' access to the virtual campus once the program is completed.**
3. **Unlimited news updates.**
4. **Unlimited access to webinars.**
5. **15 % discount for MIT Professional Education** online programs and short duration in-person programs.
6. **Exclusive announcements** of new courses, programs, and events through our **virtual Alumni Campus: Professional Network.**
7. Invitation to the **MIT Professional Education Alumni LinkedIn group.**
8. **Networking opportunities** with other MIT Professional Education Alumni.
9. In addition to **academic facilitator,** you'll be assigned a coach.

REGISTER

PROFESSIONAL EDUCATION
Digital Plus+ Programs

Are you ready to champion digital transformation in your organization, industry, and the world?

Frequently Asked Questions

global alumni
-excellence or nothing-

For more information, please contact us:

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Email: **professionalprograms@mit.edu**

Website: **https://professionalprograms.mit.edu**